

Multidisciplinary Tumor Board Summary & Outpatient Plan

Encounter Date: 21.08.2023

Referring: Dr. Priyanka Shah (GP)

Patient: Thomas Carter (25.11.53)

Primary Dx: AML, **ELN 2022 Intermediate**, **NPM1 wild-type**, **FLT3-ITD positive**

First-line therapy: **Azacitidine + venetoclax**, start **23.07.2021** (ven ramp then D1–21 each cycle). Achieved hematologic improvement with reduced transfusion needs through early 2022.

Disease Course to Date

- **Diagnosis (16.07.2021):** Presented with fatigue, gingival bleeding, and infections. Labs: WBC $18.4 \times 10^9/L$ (blasts 30%), Hgb 8.6 g/dL, Plt $62 \times 10^9/L$. Marrow: 58% blasts; **FLT3-ITD** detected (AR 0.7); **NPM1** wild type; normal karyotype. Baseline echo LVEF 60%.
 - **1L ven/aza (23.07.2021 → mid-2022):** Initial cytopenic nadirs but then **transfusion independence** by cycle 4; peripheral blasts cleared; marrow (12.2021) with **blasts 3–4%** (CRi).
 - **Clinical decline (Autumn 2022):** Gradual drop in counts, LDH rise, and small population of circulating blasts. **Bone marrow 21.09.2022** demonstrated **relapsed disease with 28% blasts**; FLT3-ITD persistent. At this time he had **ECOG 2**, CKD stage 3a, and ischemic heart disease (see comorbidities), making intensive salvage less suitable.
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Second-Line / Bridge Strategy Implemented

- **FLT3 inhibitor (Gilteritinib 120 mg PO daily)** initiated **05.10.2022**. Drug-drug interactions reviewed; azoles avoided; EKG monitoring instituted.
- **Response:** By **Nov–Dec 2022**, peripheral blasts cleared; marrow **12.12.2022: CRi** with measurable residual disease low-positive by FLT3 PCR (near limit of detection). Platelets $65–90 \times 10^9/L$; ANC $\sim 0.9–1.1 \times 10^9/L$.
- **Supportive care:** Intermittent **G-CSF** post-nadir; levofloxacin/acyclovir prophylaxis during ANC < 0.5 .
- **Adverse effects:** Transient grade 2 transaminitis (ALT peak 96 U/L) and low-grade diarrhea; QTc 456–464 ms without arrhythmia.

Given partial hematologic recovery but persistent MRD and comorbidities, the MDT favored **reduced-intensity allogeneic HSCT** if a donor was available.

Transplant Evaluation

- **HLA typing** revealed a **10/10 matched unrelated donor** available.
- **Pre-transplant work-up (June–July 2023):**
 - **Cardiology:** Hx **ischemic cardiomyopathy** with prior RCA stent (2016); LVEF **48–50%**; on metoprolol and low-dose ACE-i. Cardiologist cleared for RIC.
 - **Pulmonary:** Mild COPD (FEV₁ 72% predicted); ex-smoker (30 pack-years, quit 2016).
 - **Renal:** CKD **stage 3a** (Cr 1.4–1.6 mg/dL).
 - **Infectious disease:** Serologies negative; vaccinations updated.

Decision: Proceed to **RIC allo-HSCT** if disease remains controlled on gilteritinib; otherwise consider **FLAG-IDA + midostaurin** or **gilteritinib + azacitidine** to deepen remission first.

Current Clinic Visit (21.08.2023)

Symptoms: Improved stamina, no fevers or bleeding, occasional exertional dyspnea on hills.

Exam: Afebrile; BP 122/68; HR 74; SpO₂ 97% RA; no lymphadenopathy; lungs clear; abdomen soft; no HSM; ecchymoses resolving; ECOG **1**.

Labs: WBC **3.8×10⁹/L** (ANC 1.4), Hgb **10.8 g/dL**, Plt **92×10⁹/L**; MCV 95; **Cr 1.48 mg/dL**, AST 32, ALT 38; **LDH 220 U/L**.

MRD: FLT3-ITD **minimal residual** by highly sensitive assay; NPM1 not applicable (wt).

ECG: QTcF **461 ms** (stable).

CMV/EBV PCR: negative.

Comorbidities & Medications

- **Ischemic cardiomyopathy** (LVEF ~50%), **CAD** s/p PCI (2016)
— **metoprolol succinate 50 mg qd**, **lisinopril 5 mg qd**, **atorvastatin 40 mg qHS**.
- **Type 2 diabetes mellitus** — **metformin 500 mg bid** (held if contrast used); A1c 7.1% (07.2023).
- **COPD (mild)** — **tiotropium** daily; **albuterol** PRN.
- **GERD** — **pantoprazole 40 mg qAM**.

- **Antiplatelet/anticoagulation: Aspirin 81 mg** only when platelets $>50 \times 10^9/L$ (otherwise held). No current anticoagulant.

Allergies: none known.

MDT Recommendations & Plan

1. **Proceed toward RIC allo-HSCT** using matched unrelated donor, targeting **conditioning within 4–6 weeks** contingent on:
 - Maintaining disease control on **gilteritinib** (continue 120 mg daily up to day -2 per institutional protocol).
 - **Platelets $>50 \times 10^9/L$** and no active infection.
 - Cardio-pulmonary optimization (cardiac rehab light program; strict BP/DM control).
2. **Bridging therapy:** Continue **gilteritinib**; if cytopenias worsen or MRD rises, add **azacitidine 75 mg/m² D1–7 q28d** (dose-attenuated for renal function) while proceeding to transplant.
3. **Prophylaxis/Supportive Care:**
 - **Antimicrobials** during neutropenia: levofloxacin + acyclovir; consider mold-active azole if ANC expected <0.5 for >7 –10 days—**with QTc and gilteritinib interaction monitoring** (dose adjust/hold as needed).
 - **Transfusion thresholds:** Hgb ≥ 8 g/dL; platelets $\geq 10 \times 10^9/L$ (≥ 20 if bleeding or invasive procedures).
 - **Growth factor** PRN post-cycle.
4. **Cardiac/renal:** Continue ACE-i and beta-blocker; avoid nephrotoxins; pre-hydrate before contrast; monitor electrolytes (maintain $K^+ >4.0$, $Mg^{2+} >2.0$ for QTc).
5. **Goals of care:** Curative intent with transplant; patient understands risks (GVHD, infection, relapse, TRM) and has signed consent.
6. **Follow-up:** Weekly CBC/CMP while on gilteritinib; MRD check in 4 weeks; BMT coordinator to schedule donor work-up and tentative admission date.

Summary: Mr. Carter relapsed after venetoclax/azacitidine but **responded to gilteritinib** with CRi and low-level FLT3 MRD. Given comorbidities, a **reduced-intensity allogeneic transplant** is planned as the best chance for durable disease control.

Signed on behalf of the MDT,

[Name], MD (Hematology) • [Name], PharmD (Oncology Pharmacy) • [Name], RN (BMT Coordinator)

